

- $f(n) = O(g(n))$: there exist constants c, n_0 such that $f(n) \leq cg(n)$ for all $n \geq n_0$.
- $f(n) = \Omega(g(n))$: there exist constants c, n_0 such that $f(n) \geq cg(n)$ for all $n \geq n_0$.
- $f(n) = \Theta(g(n))$: both Big-O and Big-Omega hold.
- Master Theorem form: $T(n) = aT(n/b) + O(n^d)$, compare a to b^d .
- MergeSort: $M(n) = 2M(n/2) + O(n)$, recursion tree has $O(\log n)$ levels and $O(n)$ work per level, so $O(n \log n)$.